

T2A: A Novel Text2ASMR Framework for Generating Autonomous Sensory Meridian Response Audio

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Abstract — This research proposes T2A, an innovative neural network framework for generating Autonomous Sensory Meridian Response (ASMR) audio directly from plain text input. Unlike existing approaches, T2A does not require intermediate representations like mel-spectrograms or phonetic transcriptions. Instead, it employs a Diffusion-based UNet end-to-end network capable of zero-shot learning to produce high-fidelity binaural ASMR audio containing both speech and non-speech elements. The framework leverages BERT for text encoding and integrates a Denoising Diffusion Probabilistic Model to optimise the noise schedule, enhancing computational efficiency. A key innovation is the incorporation of feature-wise linear modulation (FiLM) layers and a conditioning network that effectively map text inputs to audio outputs, enabling versatile adaptation across contexts. Evaluations on a custom dataset of over 2000 ASMR recordings show T2A achieves promising performance, with reasonable scores on acoustic metrics like mel-cepstral distortion, log spectral distance, and F0 root mean squared error. The model exhibits strong zero-shot voice cloning capabilities and aptly captures nuanced non-speech sounds like brushing, tapping, and crinkling. This work paves the way for numerous applications, including digital foley artistry, audio post-production, and multilingual ASMR content generation. Future directions involve extending T2A to support multiple languages and further fine tuning for highly specialised audio synthesis tasks.

Impact Statement — In the past seven years, groundbreaking AI models like OpenAI's ChatGPT, Google's Gemini, and Anthropic's Claude have catalysed a transformative shift across Generative AI. Similar to the concept of a von Neumann probe, AI is now utilised to construct further iterations of AI, sparking unprecedented growth in the market for such products. Notably, open-source AI models are readily accessible on platforms like HuggingFace, democratising AI development for public use. This proliferation of AI capabilities has led developers to explore diverse applications across industries. Capitalising on this momentum, we have developed a Text-to-ASMR framework tailored to generate limitless ASMR content.

Index Terms — ASMR, Autonomous Sensory Meridian Response, Deep Learning Framework, Diffusion, Generative AI, Text-to-Speech, TTS

I. INTRODUCTION

The chronicle of Text-to-Speech (TTS) innovation unfolded with Bell Labs' AUDREY in 1952, heralding the dawn of

computational speech recognition, while the advent of articulatory synthesisers in the 1970s brought forth a deeper understanding of vocal tract mechanics[1], [2]. Later on, deep neural networks such as the Long Short Term Memory (LSTM) and Bidirectional LSTM (BiLSTM) became popular as they gained traction for higher accuracy and efficiency in generating unseen sequences of audio[3], [4].

However, it was the advent of deep learning techniques that catalysed a paradigm shift in TTS research. Models like Tacotron, introduced in 2017, and WaveNet, developed by DeepMind in 2016, revolutionised speech synthesis by leveraging neural networks to generate raw audio waveforms directly from text inputs. The subsequent refinement of WaveNet into GlowTTS, characterised by its ability to generate high-fidelity, expressive speech with improved efficiency, underscored the relentless pursuit of realism in synthetic speech generation[5], [6], [7].

Autonomous Sensory Meridian Response (ASMR), characterised by tingling sensations and relaxation induced by specific auditory or visual triggers, has gained popularity in recent years. ASMR artists, also known as ASMRtists, seek immersive experiences to evoke these sensations. ASMR first made its way onto the popular media streaming platform YouTube many years back, hallmarked by its signature near-ear sensational and soothing triggers. ASMR encompasses various triggers, ranging from interpersonal cues like whispering and personal attention (e.g., haircuts) to abstract, non-person-centric stimuli such as crisp sounds and focused tasks (e.g., towel folding, painting nails). It is important to acknowledge that this list is incomplete, and the effectiveness of specific triggers can vary significantly among individuals capable of experiencing ASMR.[8], [9], [10]

Motivated by this interest, our research introduces Text2ASMR (T2A), a framework generating ASMR speech and non-speech elements directly from plain text inputs. Unlike conventional TTS systems, T2A utilises a BERT-represented Diffusion-based UNet end-to-end network, seamlessly integrating ASMR elements without compromising fidelity or naturalness. Additionally, T2A incorporates feature-wise linear modulation (FiLM) and a general conditioning layer, enhancing its versatility and adaptability to various contexts. Furthermore, T2A adopts a specific type of Diffusion model, that is the Denoising Diffusion Probabilistic Model (DDPM), which provides a linear noise schedule to improve computational efficiency. [10], [11], [12], [13], [14], [15]

II. LITERATURE REVIEW

The literature review will cover the following works:

1. Research and/or studies oriented towards mental well-being through audio and audiovisual ASMR stimulation
2. Existing Deep Learning techniques for Text-to-Speech
3. AI-based content creation frameworks for ASMR

ASMR has gained popularity among young people for its reported ability to induce relaxation, improve sleep, reduce stress, and alleviate anxiety. While tingling is a defining feature of ASMR, it's noted that even without experiencing tingling, individuals still find ASMR beneficial for relaxation. Research has indicated that ASMR videos activate brain regions associated with reward, social bonding, and emotional arousal, reduction in depression and insomnia, suggesting potential mental health benefits. Physiological responses, such as increased skin conductance and decreased heart rate, have also been observed during ASMR experiences. Previous studies aimed to directly compare the effects of audiovisual and auditory stimulation on brain function using functional magnetic resonance imaging (fMRI). Distinct patterns of brain activation were recorded for audiovisual and auditory stimulation of ASMR. Audiovisual stimulation activated the visual cortex, middle frontal gyrus, and nucleus accumbens, while auditory stimulation activated the auditory cortex and insular cortex.[16], [17], [18]

Existing deep learning techniques for text-to-speech (TTS) synthesis, such as E3 TTS, EfficientTTS, FastDiff, Grad-TTS, and various versions of Guided-TTS, follow a similar Diffusion framework for end-to-end text-to-speech generation. These models are typically trained on widely available datasets like LJSpeech. Prior to the emergence of Diffusion models, Generative Adversarial Networks (GANs), such as HiFi-GAN, and hybrid Diffusion-GAN frameworks, like DiffGAN-TTS, were the leading approaches in this field. On a side note, Meta-StyleSpeech flexes its ability to clone unseen voices from very few audio samples, a feat that enables it to adapt to any speaker's voice. The primary advantage of such frameworks lies in their high zero-shot learning capabilities, enabling them to generalise effectively to unseen data. Diffusion models have gained significant traction across various industrial, commercial, and open-source domains due to their remarkable performance. Their success can be attributed to their ability to capture complex data distributions and generate high-quality samples, making them suitable for a wide range of applications, including TTS synthesis.[19], [20], [21], [22], [23], [24], [25]

Previous generative research for autonomous sensory meridian response (ASMR) content creation has been explored. Notably, ASMRcade, a project that leveraged a generative adversarial network (GAN) and an interactive graphical user interface (GUI), aimed to generate realistic ASMR triggers. However, ASMRcade's capabilities were limited to generating triggers, excluding speech synthesis. On the other hand, Suno's Bark, a transformer-based text-to-speech model, has demonstrated the ability to generate highly realistic, multilingual speech as well as other audio elements, such as music, background noise, and simple sound effects. From this perspective, models capable of generating non-speech audio elements could be compared to a

realistic digital emulation of a foley artist, whose role is to recreate everyday sounds for multimedia productions. [26], [27]

III. DRAWBACKS WITH TTS

1. Existing text-to-speech (TTS) models often struggle to capture a wide range of prosodic and acoustic parameters, such as pitch, tone, and loudness. Many previously encountered models tend to generalise to the average of these parameters, resulting in a narrow range of synthesised voices.
2. Prior to training, datasets must adhere to a standardised format. Benchmark datasets like LibriTTS (derived from LibriSpeech) and LJSpeech follow complex structures that may not be suitable for all frameworks. Additionally, a mapping from graphemes (smallest units of writing) to corresponding phonemes (smallest units of sound) must be established, which varies across languages, leading to a predominance of English language corpora.[28], [29]
3. With the exception of Suno's Bark, current models fail to capture interjection non-speech elements, such as exclamations like "Oh!", "Ah!", "Wow!", "Oops!", or "Hurray!", as they are often treated as noise. In contrast, T2A incorporates such instances in the dataset to feature trigger sounds and non-speech elements.

IV. METHODOLOGY

A. Data Preprocessing

Our dataset for training the T2A model included 2208 one-minute audio recordings of ASMRtists. The procedure was carried out as follows:

1. Video content from 12 different ASMRtists were collected and downloaded from YouTube
2. The video files were then converted to mono audio WAV audio format, with a sampling rate of 22,050 Hz, ensuring high-quality audio representation.
3. The audio files were trimmed down to one minute chunks.
4. To obtain the corresponding text transcripts for each individual audio sample, speech-to-text conversion was performed by an API.
5. The scripts were cleaned of non-English characters, trailing whitespaces, carriage returns, etc.
6. Interjections such as exclamation, questioning, pause were represented by '!', '?', '-', ellipsis and many more.
7. Trigger sounds such as brushing, rustling, tapping, crinkling, etc were represented by [brushing], [rustling], [tapping], [crinkling] and so on.
8. To modulate the intensity of the non-speech elements, prefix tags were introduced, enabling granular control over the synthesis. For instance, [mild], [soft], [vigorous], and [loud] could precede the aforementioned trigger sound tags, providing additional contextual information for the model to generate nuanced auditory outputs.

B. Model Architecture

A sequence of tokenizers, encoder-decoder networks and CNNs together to make the training architecture below (see Figure 1):

1. *Text Input*: The framework starts with a text input, which can be a sentence or a paragraph describing the desired audio.
2. *Bert Tokenizer*: The text input is tokenized using the BertTokenizer from the Hugging Face Transformers library.
3. *Bert Model*: The tokenized text is passed through the BERT Model, which generates contextualised text embeddings.
4. *Conditioning Network*: The text embeddings from the BERT Model are passed through this network, which produces conditioned features.
5. *Diffusion (noising)*: The Diffusion model generates an initial noisy waveform by adding noise to the corresponding clean waveform associated with that text input (see Figure 2).
6. *U-Net with FiLM*: The conditioned features from the Conditioning Network and the initial noisy waveform are fed into the U-Net with FiLM layers. The U-Net applies downsampling and upsampling operations (see Figure 3), along with FiLM layers (see Figure 4) that modulate the feature maps based on the conditioned features. The output is predicted noise.
7. *Waveform update (denoising)*: The Diffusion Model takes the output from the U-Net with FiLM and performs the diffusion process. It then denoises the noisy waveform by removing some of the added noise in that timestep.
8. *Iterative Process*: Steps 6-8 are repeated iteratively for multiple timesteps until the waveform with near-zero noise is achieved.. At each timestep, the Diffusion model generates a noisy waveform, passes it to the UNet-FiLM network and cleans the waveform with the predicted noise.
9. *Waveform Output*: The final output waveform after ‘t’ iterative denoising schedules is the desired audio.

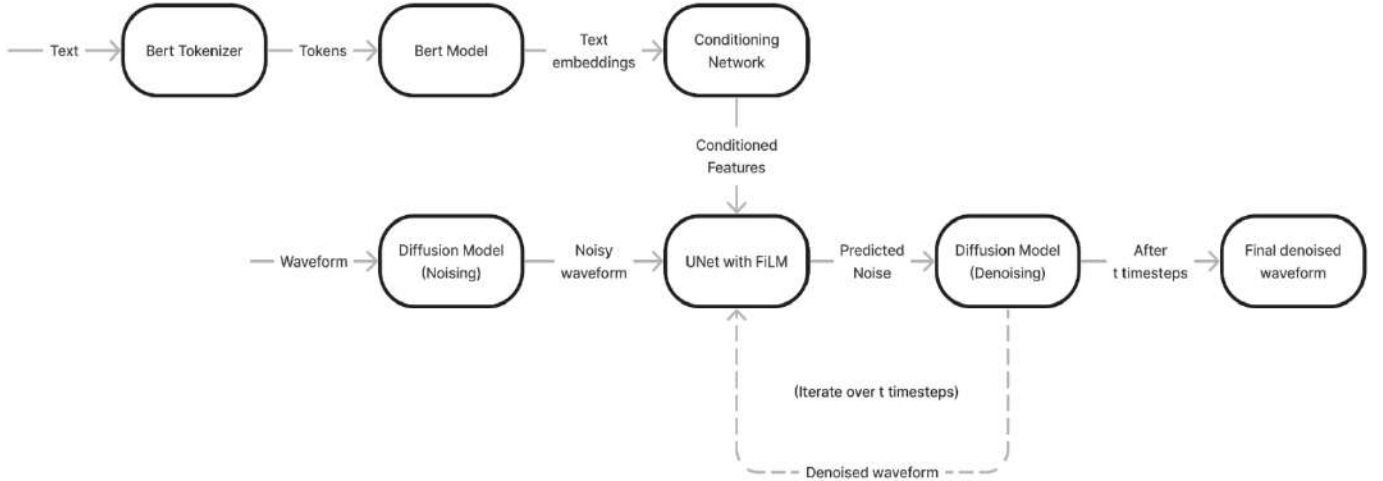


Fig. 1: Representation of the model architecture for training

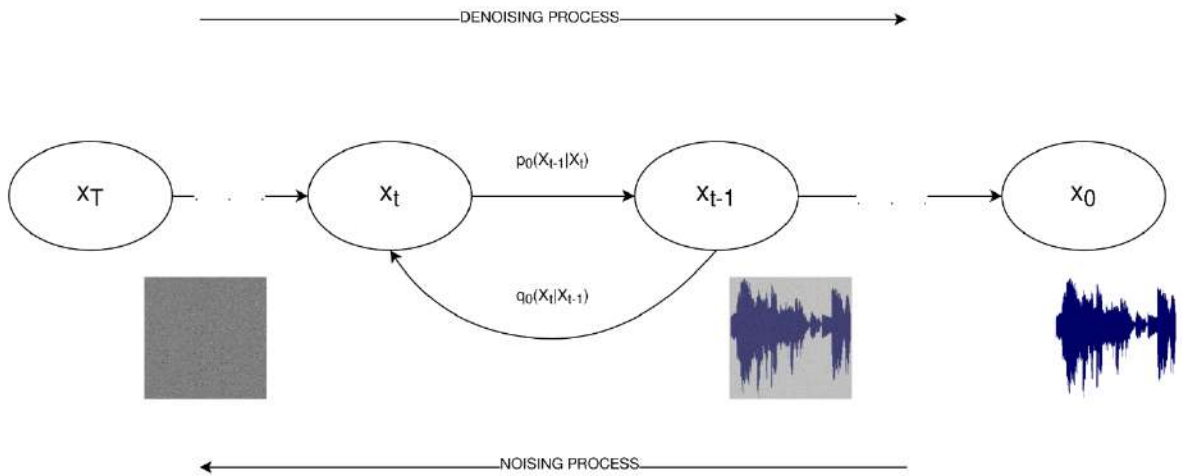


Fig. 2: Working of DDPM noising and denoising schedules

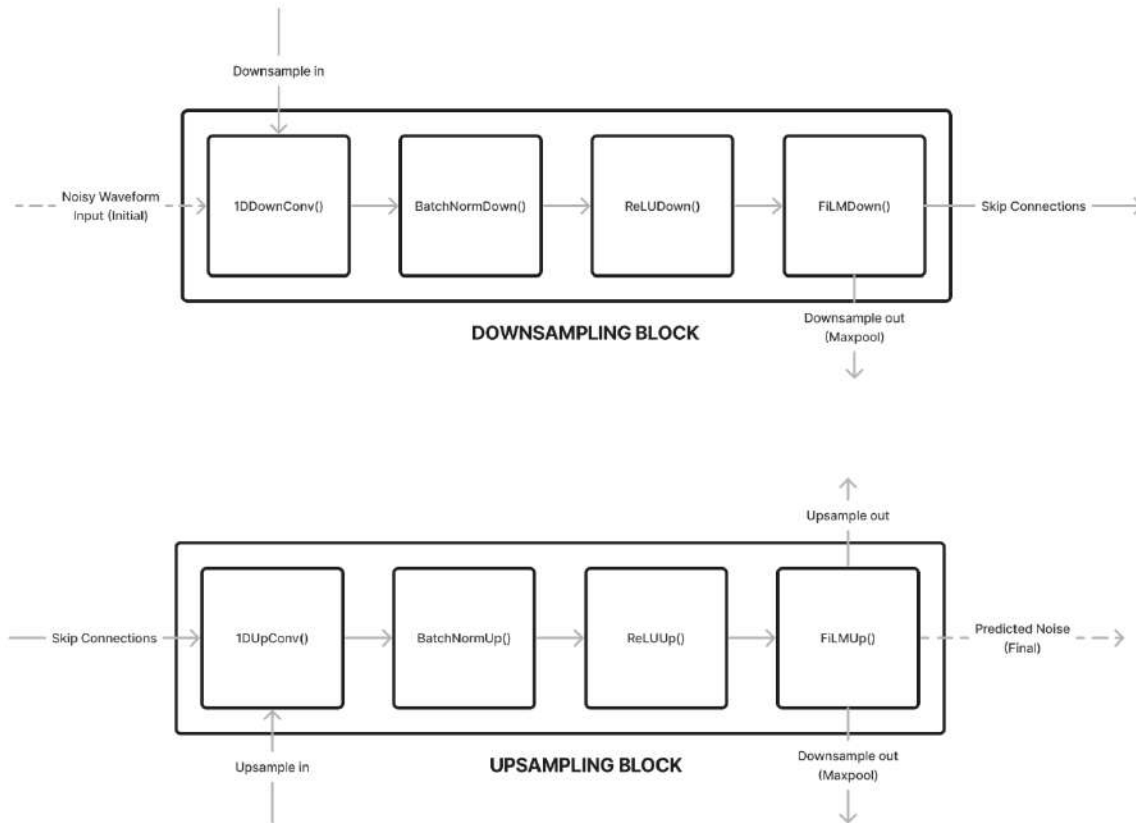


Fig. 3: Modified Upsampling and Downsampling Blocks of the UNet-FiLM network

Initially, the diffusion process introduces noise to the waveforms, which are then passed through the UNet's upsampling and downsampling blocks. As depicted in Figure 3, the T2A model slightly modifies these blocks by incorporating a FiLM layer to integrate conditioned features from text inputs (see Figure 4). The FiLM layer's advantage lies in its ability to perform an elegant mapping of text input to audio output (the noisy waveform in this case), implicitly creating intermediate representations that the upsampling blocks can subsequently learn more efficiently. The mapping element is crucial, and the Condition Network proved to be the best-performing candidate among others, such as Attention and Gated Biases.

From a technical standpoint, the waveform and text input can be represented as one-dimensional vectors of audio frames and tokenized embeddings, respectively. These frames are arranged sequentially, but during the FiLM mapping, the textual embeddings are mapped to each segment of the audio vector in parallel, significantly improving processing time. Before entering the FiLM layer, the text input undergoes shifting and scaling transformations (see Figure 4) to better align with the noisy waveform. The output of the FiLM layer is another one-dimensional vector that flows into the subsequent downsampling/upsampling blocks.

The UNet's skip connections are another notable feature, enabling intermediate evaluation and correction of the network's weights by allowing it to look back in time. This mechanism has facilitated faster training and smoother gradient descent for the UNet compared to other models.

Moreover, the FiLM layers enable zero-shot voice cloning, a remarkable capability where a small segment of audio from an individual can be leveraged to replicate an entire ASMR conversation in their voice. This property holds significant potential for applications such as audiobook narration, virtual assistants, and speech synthesis.

The FiLM layers' potential for zero-shot voice cloning and Foley sound generation exemplifies the mapping mechanism in integrating conditioned features from text inputs and audio outputs. Despite the relatively infrequent occurrence of non-speech sounds in the dataset, the network has remarkably captured their nuances and performs exceptionally well. This versatility broadens the potential applications of the model, making it suitable for tasks such as audio post-production, sound design, and multimedia content creation.

The complete work can be found here: <https://huggingface.co/datasets/aoxo/text2asmr-uncensored>

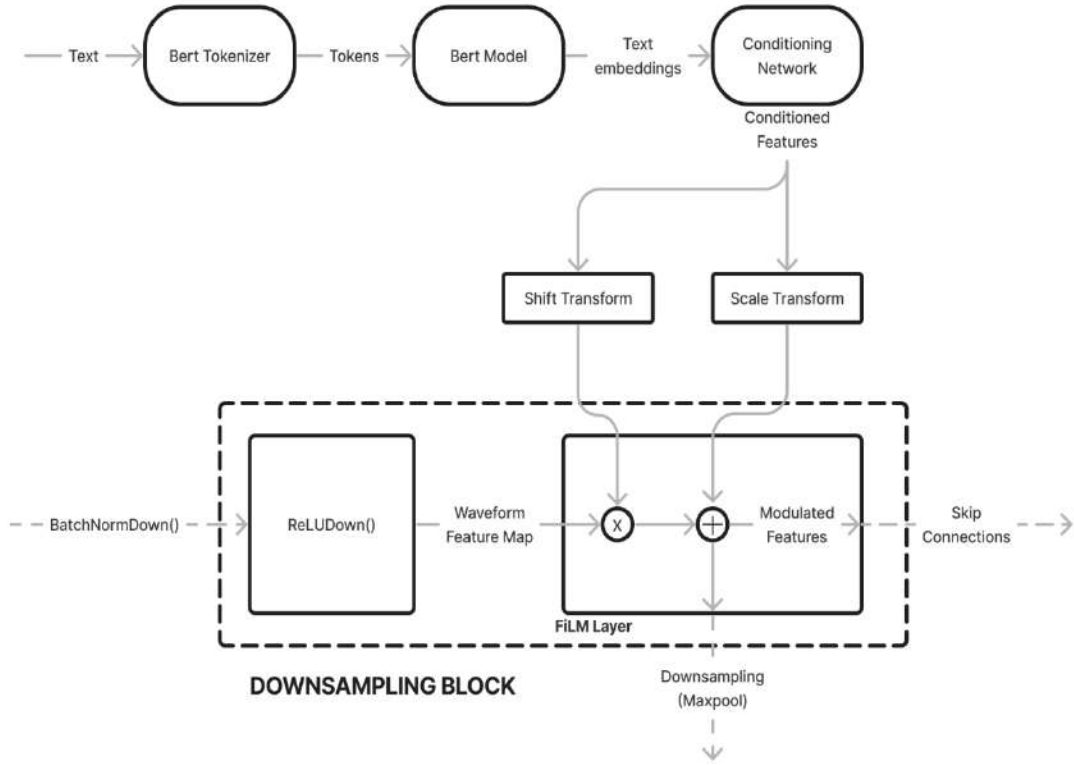


Fig. 4: Working of Conditioning Network and Feature-wise Linear Modulation Layer (FiLM)

V. RESULTS AND DISCUSSIONS

The model was trained for a total of 10 epochs and 1000 denoising timesteps per epoch. We have considered three acoustic metrics:

A. Mel-cepstral Distortion (MCD):

MCD measures the spectral distortion between the generated and reference audio by computing the Euclidean distance between the Mel-cepstral coefficients of the two audio signals.[30] Parameters: 25 Mel Frequency Cepstral Coefficients (MFCC) with frame step size of 5ms.

$$MCD = \frac{10\sqrt{2}}{\ln 10} \frac{1}{T} \sum_{t=1}^T \sqrt{\sum_i (c_{ti} - \hat{c}_{ti})^2}$$

where, c_{ti} and $\hat{c}_{ti}$ are the reference and generated waveform Mel-cepstral coefficients, respectively.

B. Log Spectral Distance (LSD):

Similar to MCD, LSD measures the spectral distortion between the generated and reference waveforms by computing the log-spectral distance between their magnitude spectra.[31]

$$LSD = \sqrt{\left(\frac{1}{N}\right) \sum (\log(s_t) - \log(s_r))^2}$$

where,

s_t - generated waveform magnitude spectrum

s_r - reference waveform magnitude spectra.

N- number of frequency bins

C. F0 Root Mean Squared Error (F0 RMSE):

This metric evaluates the prosodic accuracy of the predicted fundamental frequency (F0) compared to the reference contour.[32]

$$F0 \text{ RMSE} = \sqrt{\frac{1}{T} \sum (fo_t - fo_r)^2}$$

where: fo_t - predicted F0 contours

fo_r - reference F0 contours

T - no of frames

Metric	Score
MCD	4.433
LSD	1.819
F0 RMSE	10.4 Hz
MSE	0.0027%
RMSE	0.0217%
MAE	0.0018%

Table 1: Summary of metric scores for T2A

VI. CONCLUSION AND FUTURE SCOPE

We have proposed a novel Text2ASMR framework, T2A, built upon Diffusion mechanism, UNet framework, FiLM layers and Conditioning Networks, that is effective at generating ASMR audio from a textual input. T2A displays its prowess at zero-shot adaptability to a wide range of unseen voices and content. It is also capable of prolonged speech generation greater than one minute by stitching together previous chunks using the inbuilt attention mechanism in UNet.

As part of future work direction, we plan to extend T2A to support multilingual ASMR content generation as well as finetune a parallel model as a digital clone of a foley artist.

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